KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

EMPLOYEE PERFORMANCE EVALUATION AND RATING SYSTEM

Submitted by

VIJAYAKUMARI B

25BIT162

Under the Guidance of

MUGILAN R

Technical Trainer

Technical Department, KG College of Arts and Science

Academic Year: **2025 – 2026**

APRIL 2026

Semester: **I**

DECLARATION

I hereby declare that the project entitled "**EMPLOYEE PERFORMANCE EVALUATION AND RATING SYSTEM"** submitted in partial fulfillment of the requirements for the course [**JAVA PROGRAMMING]** is a **record of my original work** carried out under the guidance of **[MUGILAN R]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"EMPLOYEE PERFORMANCE EVALUATION AND RATING SYSTEM"** is a **bona fide work** carried out by **[VIJAYAKUMARI B] [B.SC.IT "B" (2025-2028)]**, **[B.SC.INFORMATION AND TECHNOLOGY]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

Head of the Department
(Technical Head, KGCAS)

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The **Employee Performance Evaluation and Rating System** is a Java-based application designed to automate the process of evaluating employee performance in an organization. In many organizations, performance evaluation is carried out manually, which can lead to calculation errors, inconsistency in ratings, and difficulty in maintaining proper records. This project aims to simplify the evaluation process by providing an automated system that calculates performance scores and assigns ratings accurately.

The system allows users to enter employee details such as employee ID, name, attendance, project performance, teamwork, and communication skills. Based on these inputs, the system calculates the overall performance score using a weighted average method and assigns a rating such as *Excellent, Good, Average,* or *Needs Improvement*.

The application is developed using Core Java programming concepts including classes, objects, methods, loops, conditional statements, and collections like ArrayList. The system is implemented as a console-based application and can be further enhanced using Java Swing for a graphical user interface.

The main objective of the system is to improve evaluation efficiency, reduce human errors, and maintain organized employee performance records. This project demonstrates the practical implementation of object-oriented programming concepts in Java and provides a foundation for developing more advanced employee management systems in the future.

1.INTRODUCTION

In today's competitive business environment, organizations need an efficient system to evaluate employee performance accurately and fairly. Traditionally, employee performance evaluation is done manually using paper-based records or spreadsheets, which can be time-consuming, error-prone, and difficult to manage.

With the advancement of technology, automated systems have become essential for improving the efficiency and reliability of performance evaluation processes. A computerized system helps in storing employee data, calculating performance scores, and generating ratings quickly and accurately.

The **Employee Performance Evaluation and Rating System** is a Java-based application developed to automate the evaluation process of employees. This system allows the user to enter employee details such as ID, name, attendance, project performance, teamwork, and communication skills. Based on these inputs, the system calculates a total performance score using a weighted average method and assigns a rating such as *Excellent, Good, Average,* or *Needs Improvement*.

The application is designed using Core Java concepts including classes, objects, methods, loops, conditional statements, and ArrayList for dynamic data storage. It follows a menu-driven approach, making it user-friendly and easy to operate.

The main objective of this system is to reduce manual effort, ensure accurate performance evaluation, maintain organized employee records, and demonstrate the practical application of Java programming in real-world scenarios.

1.1 Objective

The main objective of the **Employee Performance Evaluation and Rating System** is to develop a simple and efficient application using Core Java that automates the process of evaluating employee performance and assigning ratings.

The specific objectives of this project are:

- To design and develop a Java-based employee evaluation system.

- To store and manage employee details such as ID, name, and performance parameters.
- To automate the calculation of employee performance scores using a weighted average method.
- To assign performance ratings such as *Excellent, Good, Average,* and *Needs Improvement* based on the calculated score.
- To reduce manual errors in evaluating employee performance.
- To provide a user-friendly, menu-driven system for easy interaction
- To enable searching and displaying employee performance details efficiently To demonstrate the implementation of Core Java concepts such as classes, objects, methods, loops, conditional statements, and ArrayList.

1.2 Overview

The **Employee Performance Evaluation and Rating System** is a menu-driven application developed using the Java programming language. The system provides a digital platform to manage employee details and evaluate their performance efficiently.

The application allows the user to enter employee information such as employee ID, name, attendance, project performance, teamwork, and communication skills. Based on these inputs, the system calculates the total performance score using a weighted average method.

The program automatically assigns a performance rating such as *Excellent, Good, Average,* or *Needs Improvement* according to the calculated score. It also allows users to display all employee details and search for a specific employee using their ID.

This system simplifies the evaluation process by eliminating manual calculations and reducing errors. It is implemented as a console-based application using Core Java concepts such as classes, objects, methods, loops, conditional statements, and ArrayList.

This project demonstrates the practical application of Java programming in developing real-world systems for employee management and performance analysis.

1.3 Features

The **Employee Performance Evaluation and Rating System** includes the following main features:

- **Employee Data Entry** – Allows the user to enter employee details such as ID, name, and performance parameters.
- **Performance Calculation** – Calculates the total performance score automatically using weighted criteria like attendance, project score, teamwork, and communication.

- **Rating System** – Assigns performance ratings such as *Excellent, Good, Average,* and *Needs Improvement* based on the calculated score.
- **Display Employee Details** – Displays complete information of all employees including scores and ratings.
- **Search Functionality** – Allows the user to search for an employee using their employee ID.
- **Menu-Driven Interface** – Provides a simple and easy-to-use interface for user interaction.
- **Error Reduction** – Minimizes manual calculation errors in performance evaluation.
- **Efficient Data Management** – Uses ArrayList to store and manage employee records dynamically.

2.PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should function and what features are needed to achieve accurate and efficient results.

The analysis phase focuses on identifying the limitations of the existing system and proposing a computerized solution that improves performance evaluation and data management.

2.1 Problem Definition

In many organizations, employee performance evaluation is often carried out manually using paper records or basic tools like spreadsheets. This traditional method creates several challenges.

Some common problems include:

- Errors in performance calculation
- Time-consuming evaluation process
- Difficulty in managing multiple employee records
- Lack of proper and organized performance data
- Inconsistency in assigning ratings
- Increased workload for management

Manual evaluation systems can lead to inaccurate performance assessments and may affect decision-making processes such as promotions, rewards, and employee development.

Therefore, there is a need for an automated system that can efficiently manage employee data, calculate performance scores accurately, and assign appropriate ratings.

The **Employee Performance Evaluation and Rating System** is designed to solve these problems by providing a computerized solution that automates the evaluation process, reduces errors, and ensures faster and more reliable performance assessment.

2.2 Project Overview

The **Employee Performance Evaluation and Rating System** is a Java-based application developed to simplify and automate the process of evaluating employee performance in an organization.

The system provides a menu-driven interface where users can enter employee details such as employee ID, name, and various performance parameters including attendance, project score, teamwork, and communication skills.

Based on these inputs, the system automatically calculates the overall performance score using a weighted average method and assigns a suitable rating.

The system includes the following functionalities:

- Entering employee details and performance parameters
- Calculating overall performance score automatically
- Assigning ratings based on performance score
- Displaying all employee records with complete details
- Searching for a specific employee using employee ID

This project is implemented using Core Java programming concepts such as classes, objects, methods, loops, conditional statements, and ArrayList for dynamic data storage.

the application is designed as a console-based program and can be further enhanced using Java Swing for a graphical user interface.

2.3 Scopes and Limitations

Scope

The scope of the **Employee Performance Evaluation and Rating System** includes the following:

- Managing employee details and performance data
- Evaluating employee performance based on multiple parameters

- Calculating overall performance scores automatically
- Assigning performance ratings based on predefined criteria
- Displaying and searching employee records efficiently
- Demonstrating the practical use of Core Java programming concepts

The system can be used in:

- Small and medium-scale organizations
- Companies for employee performance tracking
- Educational institutions for student evaluation
- Training centers and project teams

Limitations

- The system is console-based and does not include a graphical user interface (GUI).
- Data is not stored permanently (no database or file storage).
- Limited to basic performance parameters (attendance, project, teamwork, communication).
- Does not support multi-user access or networking features.
- Security features such as login authentication are not included.
-

3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before implementation. In this phase, the system is divided into different modules, and the workflow is clearly defined. Proper system design helps in organizing the program efficiently and simplifies development and maintenance.

The **Employee Performance Evaluation and Rating System** is designed as a menu-driven application where users can add employee details, evaluate performance, and view results easily.

The system is divided into several modules, each responsible for performing a specific function.

3.1 Modules

The main modules of the **Employee Performance Evaluation and Rating System** are explained below:

1. Employee Data Entry Module

This module is responsible for collecting employee details and performance parameters.

Functions of this module include:

- Accepting employee ID and name
- Accepting performance inputs such as:
  - Attendance
  - Project Score
  - Teamwork
  - Communication
- Storing employee data in the system using ArrayList
- This module ensures that all necessary data is entered correctly for evaluation.

2. Performance Calculation Module

This module calculates the overall performance score of each employee.

The system uses a weighted average method:

- Attendance → 20%
- Project Score → 40%
- Teamwork → 20%
- Communication → 20%
- **Formula:**
- Total Score = (Attendance × 0.2) + (Project Score × 0.4) + (Teamwork × 0.2) + (Communication × 0.2)
- The calculation is performed automatically using Java logic.

3. Rating Assignment Module

This module assigns a rating based on the calculated performance score.

Rating Criteria:

- 85 and above → Excellent
- 70 to 84 → Good
- 50 to 69 → Average
- Below 50 → Needs Improvement
- This helps in categorizing employees based on their performance level.

4. Display Module

This module is used to display employee details along with their performance score and rating.

The displayed information includes:

- Employee ID
- Name
- Performance parameters
- Total Score
- Rating
- It allows users to view all employee records clearly.

5. Search Module

This module allows the user to search for a specific employee using their employee ID.

- Functions include:
- Accepting employee ID as input
- Searching in the employee list
- Displaying the details if found
- Showing a message if the employee is not found

6. Exit Module

This module allows the user to safely exit the application.

When the user selects the exit option, the system terminates the program execution.

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Employee Performance Evaluation and Rating System**, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the need for efficient employee performance evaluation.

The main problems identified in manual evaluation systems are:

- Manual calculations may lead to incorrect performance scores
- Evaluation process is time-consuming
- Difficulty in managing multiple employee records
- Lack of proper and organized data
- Inconsistency in assigning performance ratings

Based on these problems, the system requirements were defined to develop an automated evaluation system that simplifies employee performance assessment.

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

- Employee Data Entry Module
- Performance Calculation Module
- Rating Assignment Module

- Display Module
- Search Module
- Exit Module

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

After designing the system, the program is implemented using the Java programming language.

The following steps were followed during development:

- Creation of classes and methods
- Implementation of menu-driven logic
- Writing code for performance calculation
- Handling user input using Scanner
- Storing data using ArrayList
- Displaying employee details and ratings

Core Java concepts such as classes, objects, loops, conditional statements, methods, and collections were used during development.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

This phase ensures that the program runs without errors and produces accurate results.

EXAMPLE TEXT CASE

test case	**Expected Result**
add employee	Employee details stored correctly
Enter the performance values	values accepted properly
Calculate performance	Correct score and rating displayed
Search employee	correct employee details shown
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

After successful testing, the results of the system are documented. The system generates performance scores and ratings based on the entered employee data.

The results are demonstrated through program execution showing input and output.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

- **Algorithm for Employee Performance Evaluation System:**
- Start the program.
- Display the menu options.
- Choose an option from the menu.
- If "Add Employee":
  - Enter employee details and performance parameters.
  - Store the data.
- If "Calculate Performance":
  - Calculate total score for each employee.
  - Assign rating based on score.
- If "Display Employees":
  - Show all employee details.
- If "Search Employee":
  - Enter employee ID.
  - Display matching record.

- Repeat until user chooses exit.
- End the program.

4.7. Flowchart

Students should draw a flowchart showing the program flow.

Example Flow:

Start
↓
Display Menu
↓
Enter Choice
↓
Add / Calculate / Display / Search
↓
Show Result
↓
Repeat or Exit
↓
End

5.SOURCE CODE

```
import java.util.*;


class Employee {
    int empId;
    String name;
    double attendance;
    double projectScore;
    double teamwork;
    double communication;
```

```
double totalScore;
String rating;

Employee(int empId, String name, double attendance, double projectScore,
        double teamwork, double communication) {
    this.empId = empId;
    this.name = name;
    this.attendance = attendance;
    this.projectScore = projectScore;
    this.teamwork = teamwork;
    this.communication = communication;
}

  void calculatePerformance() {
    totalScore = (attendance * 0.2) +
                 (projectScore * 0.4) +
                 (teamwork * 0.2) +
                 (communication * 0.2);



    if (totalScore >= 85)
        rating = "Excellent";
    else if (totalScore >= 70)
        rating = "Good";
    else if (totalScore >= 50)
        rating = "Average";
    else
        rating = "Needs Improvement";
}
```

```
    void display() {
        System.out.println("\n--- Employee Details ---");
        System.out.println("ID: " + empId);
        System.out.println("Name: " + name);
        System.out.println("Attendance: " + attendance);
        System.out.println("Project Score: " + projectScore);
        System.out.println("Teamwork: " + teamwork);
        System.out.println("Communication: " + communication);
        System.out.println("Total Score: " + totalScore);
        System.out.println("Rating: " + rating);
    }
}

public class EmployeeEvaluationSystem {

    static ArrayList<Employee> empList = new ArrayList<>();

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;

        do {
            System.out.println("\n===== Employee Performance System =====");
            System.out.println("1. Add Employee");
            System.out.println("2. Calculate Performance");
            System.out.println("3. Display All Employees");
            System.out.println("4. Search Employee");
            System.out.println("5. Exit");
```

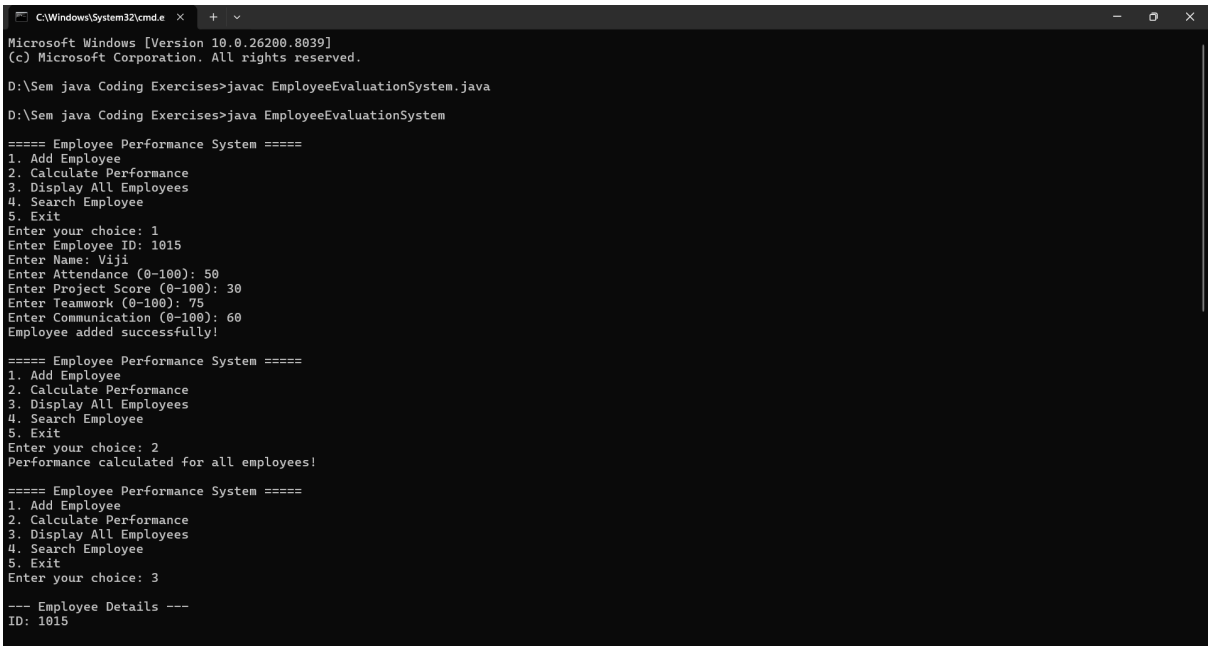
```
System.out.print("Enter your choice: ");
choice = sc.nextInt();

switch (choice) {

    case 1:
        addEmployee(sc);
        break;

    case 2:
        calculateAll();
        break;

    case 3:
        displayAll();
        break;

    case 4:
        searchEmployee(sc);
        break;

    case 5:
        System.out.println("Exiting program...");
        break;

    default:
        System.out.println("Invalid choice!");
}
```

```
    } while (choice != 5);

    sc.close();
}


static void addEmployee(Scanner sc) {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();
    sc.nextLine();

    System.out.print("Enter Name: ");
    String name = sc.nextLine();

    System.out.print("Enter Attendance (0-100): ");
    double att = sc.nextDouble();

    System.out.print("Enter Project Score (0-100): ");
    double proj = sc.nextDouble();

    System.out.print("Enter Teamwork (0-100): ");
    double team = sc.nextDouble();

    System.out.print("Enter Communication (0-100): ");
    double comm = sc.nextDouble();

    Employee emp = new Employee(id, name, att, proj, team, comm);
    empList.add(emp);
```

```
    System.out.println("Employee added successfully!");
}

   static void calculateAll() {
    if (empList.isEmpty()) {
        System.out.println("No employees found!");
        return;
    }

    for (Employee emp : empList) {
        emp.calculatePerformance();
    }

    System.out.println("Performance calculated for all employees!");
}

   static void displayAll() {
    if (empList.isEmpty()) {
        System.out.println("No employees to display!");
        return;
    }

    for (Employee emp : empList) {
        emp.display();
    }
}

  static void searchEmployee(Scanner sc) {
    System.out.print("Enter Employee ID to search: ");
```

```
        int id = sc.nextInt();


        for (Employee emp : empList) {
            if (emp.empId == id) {
                emp.display();
                return;
            }
        System.out.println("Employee not found!");
}
}
```

6.INPUT AND OUTPUT SCREENS



```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

D:\Sem java Coding Exercises>javac EmployeeEvaluationSystem.java

D:\Sem java Coding Exercises>java EmployeeEvaluationSystem

===== Employee Performance System =====
1. Add Employee
2. Calculate Performance
3. Display All Employees
4. Search Employee
5. Exit
Enter your choice: 1
Enter Employee ID: 1015
Enter Name: Viji
Enter Attendance (0-100): 50
Enter Project Score (0-100): 30
Enter Teamwork (0-100): 75
Enter Communication (0-100): 60
Employee added successfully!

===== Employee Performance System =====
1. Add Employee
2. Calculate Performance
3. Display All Employees
4. Search Employee
5. Exit
Enter your choice: 2
Performance calculated for all employees!

===== Employee Performance System =====
1. Add Employee
2. Calculate Performance
3. Display All Employees
4. Search Employee
5. Exit
Enter your choice: 3

--- Employee Details ---
ID: 1015
```

```
C:\Windows\System32\cmd.e

--- Employee Details ---
ID: 1015
Name: Viji
Attendance: 50.0
Project Score: 30.0
Teamwork: 75.0
Communication: 60.0
Total Score: 49.0
Rating: Needs Improvement

===== Employee Performance System =====
1. Add Employee
2. Calculate Performance
3. Display All Employees
4. Search Employee
5. Exit
Enter your choice: 4
Enter Employee ID to search: 1015

--- Employee Details ---
ID: 1015
Name: Viji
Attendance: 50.0
Project Score: 30.0
Teamwork: 75.0
Communication: 60.0
Total Score: 49.0
Rating: Needs Improvement

===== Employee Performance System =====
1. Add Employee
2. Calculate Performance
3. Display All Employees
4. Search Employee
5. Exit
Enter your choice: 5
Exiting program...

D:\Sem java Coding Exercises>
```

Output Explanation

- **Program Start**
  - The system displays a **menu** with options like Add Employee, Calculate Performance, Display, Search, and Exit.
- **Adding Employee (Choice 1)**
  - You entered employee details:
    - ID: 1015
    - Name: Viji
    - Performance values (Attendance, Project, Teamwork, Communication)
  - The system stores this data successfully.
- **Calculating Performance (Choice 2)**
  - The system calculates the **total score** using the formula:
    (Attendance × 0.2) + (Project × 0.4) +
    (Teamwork × 0.2) + (Communication × 0.2)

  - For your input, the total score = **49.0**
- **Displaying All Employees (Choice 3)**
  - The system shows all employee details.
  - Based on the score (49.0), the rating is:
    - **Needs Improvement**
- **Searching Employee (Choice 4)**
  - When you enter ID **1015**, the system finds and displays the same employee details.
- **Exit (Choice 5)**
  - The program stops execution safely.

7.DISCUSSION & CONCLUSION

7.1 Discussion

The **Employee Performance Evaluation and Rating System** was developed to automate the process of evaluating employee performance in an organization. The system allows users to enter employee details, calculate performance scores, and assign ratings automatically.

During the development process, Core Java programming concepts such as classes, objects, methods, loops, conditional statements, and ArrayList were applied effectively. The system successfully performs performance calculations using a weighted average method and assigns appropriate ratings based on predefined criteria.

The system was tested with different inputs to ensure accuracy and reliability. The results show that the system reduces manual effort, minimizes errors in evaluation, and improves efficiency in managing employee performance data.

7.2 Conclusion

The **Employee Performance Evaluation and Rating System** successfully demonstrates the practical application of Core Java programming concepts in developing a real-world application.

The system automates the evaluation process, ensures accurate calculation of performance scores, and provides a structured way to assign employee ratings. It simplifies the management of employee data and improves decision-making processes in organizations.

This project helps in understanding how Java can be used to build practical software systems for performance management. The system can be further enhanced by adding features such as database connectivity, graphical user interface (GUI), report generation, and multi-user access.

REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – *Java: The Complete Reference*
- E. Balagurusamy – *Programming with Java*

Websites

- https://www.oracle.com/java
- https://www.geeksforgeeks.org
- https://www.javatpoint.com
- https://www.w3schools.com/java